Applied Math 50: Overview

1. Recap and further discussion of “The World Series Competition” by Frederick Mosteller
2. Five minute break
3. Beginning tutorial in Matlab

Lecture notes will be posted on Course site

Quick detour

"We love sports, it's a great way to teach people about math and statistics."

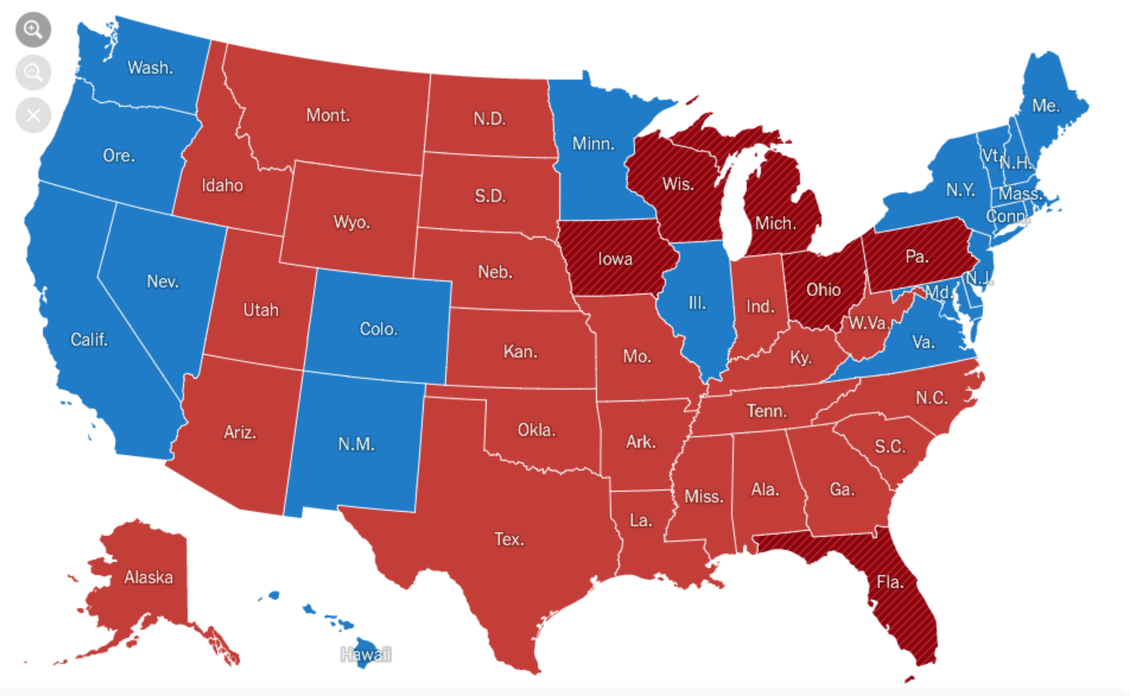
Statistician Nate Silver on
The Colbert Report, Jan 27 2014

Mathematical model of the 2012 US general election: all 50 states predicted correctly

http://www.fivethirtyeight.com/

http://fivethirtyeight.blogs.nytimes.com/

But … 2016?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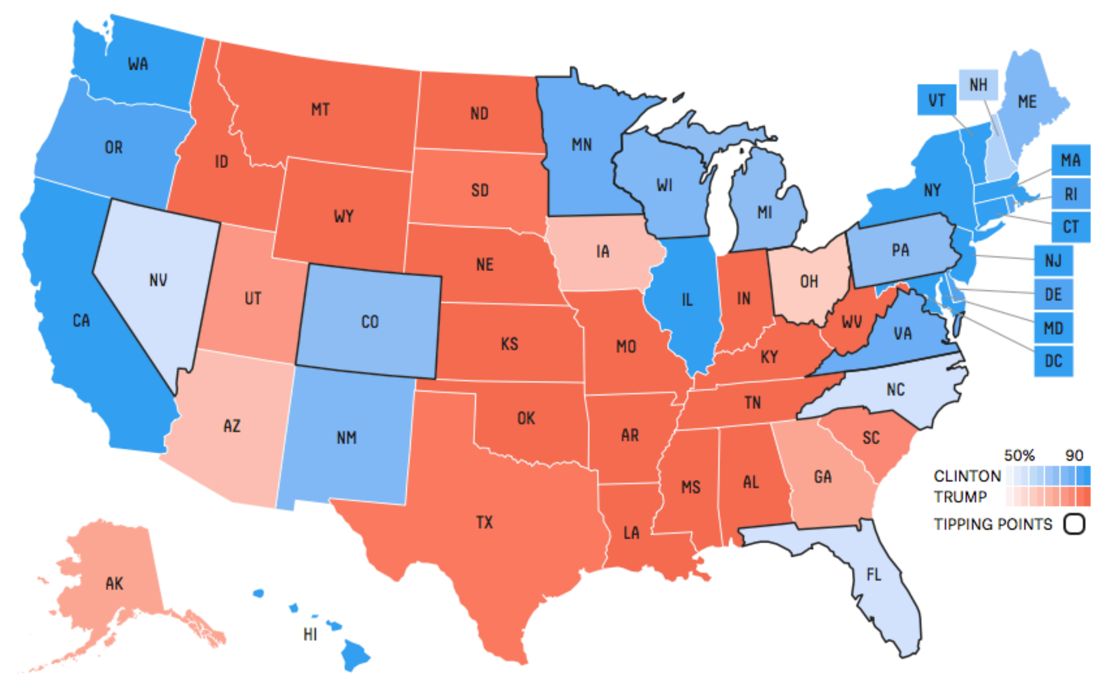
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Mosteller's World Series model

JOURNAL OF THE AMERICAN STATISTICAL ASSOCIATION

Number 259 SEPTEMBER 1952 *Volume 47*

THE WORLD SERIES COMPETITION*

FREDERICK MOSTELLER
Harvard University

Full paper available on Canvas site

Mosteller's World Series model

- Assume that in the World Series, one team wins with probability *p*, like a flipping an unfair coin
- The better team is the one with the higher probability of winning a single coin flip
- The point of playing a "series" is that you would like to amplify the chance the better team wins
- Central question: how often does the better team win?

Binomial trials

- **Suppose you are given a coin, and the probability of getting a head is *p*. In *N* trials the probability of getting exactly *m* heads is**

$$P(N,m) = \frac{N!}{(N-m)!\,m!}\,p^m(1-p)^{N-m}$$

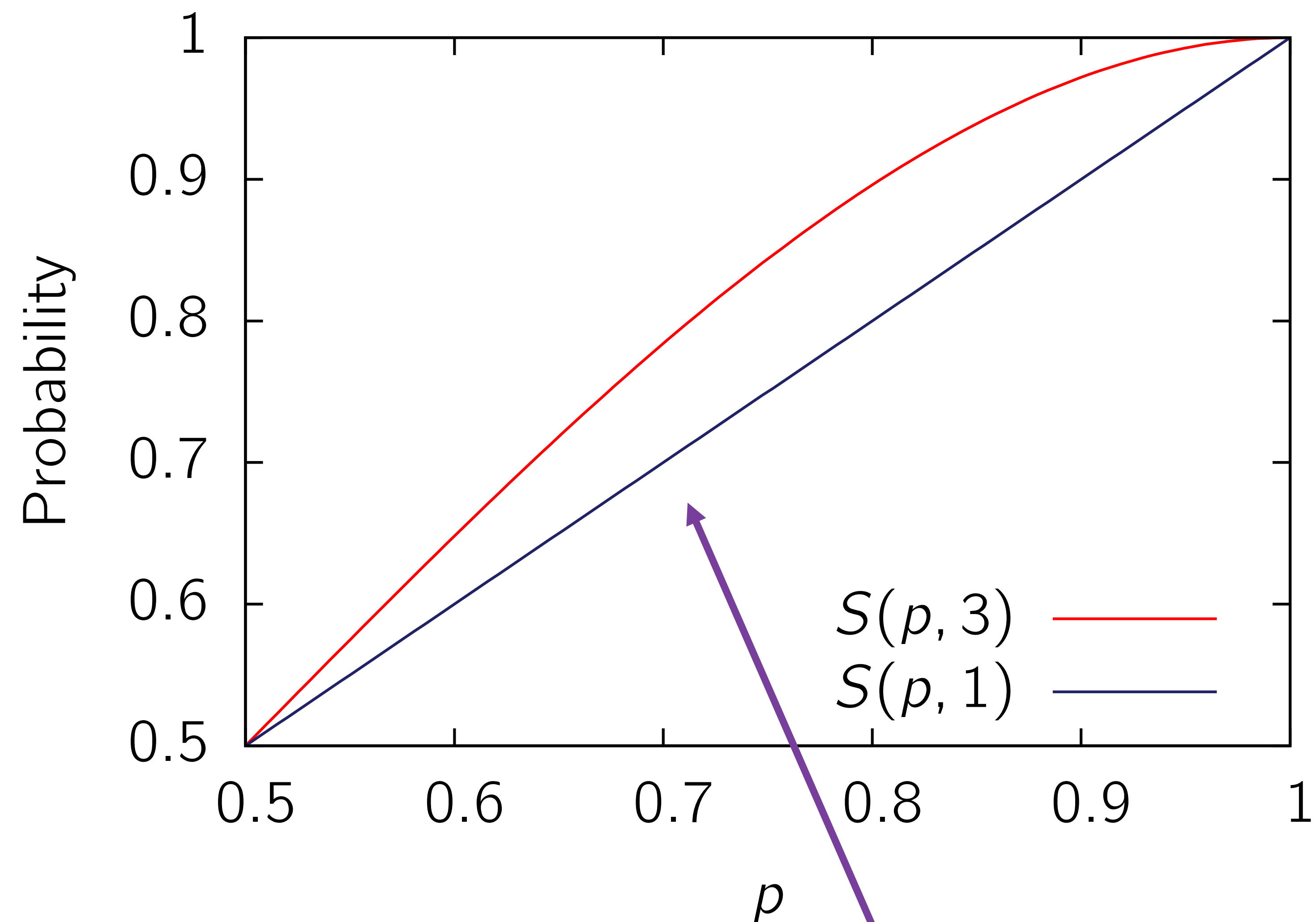
- **Alternative notation**

$$P(N,m) = \binom{N}{m} p^m(1-p)^{N-m}$$

"*N* choose *m*" : the number of ways to choose *m* items from *N* things

Success probability

- **Define *S*(*p*, *N*) to the be the probability of winning an *N* game series**



- **Then**

$$S(p,3) = \Pr(Win\ 3) + \Pr(Win\ 2)$$
$$= P(3,3) + P(3,2) = p^3 + 3p^2(1-p)$$

- **Winning combinations:**

BBB, BBW, BWB, WBB

Mosteller's notation
B: better team wins
W: worse team wins

Amplification factor

- Let ϵ be the "advantage," so that $p = \frac{1}{2} + \epsilon$
- Mosteller's calculation:

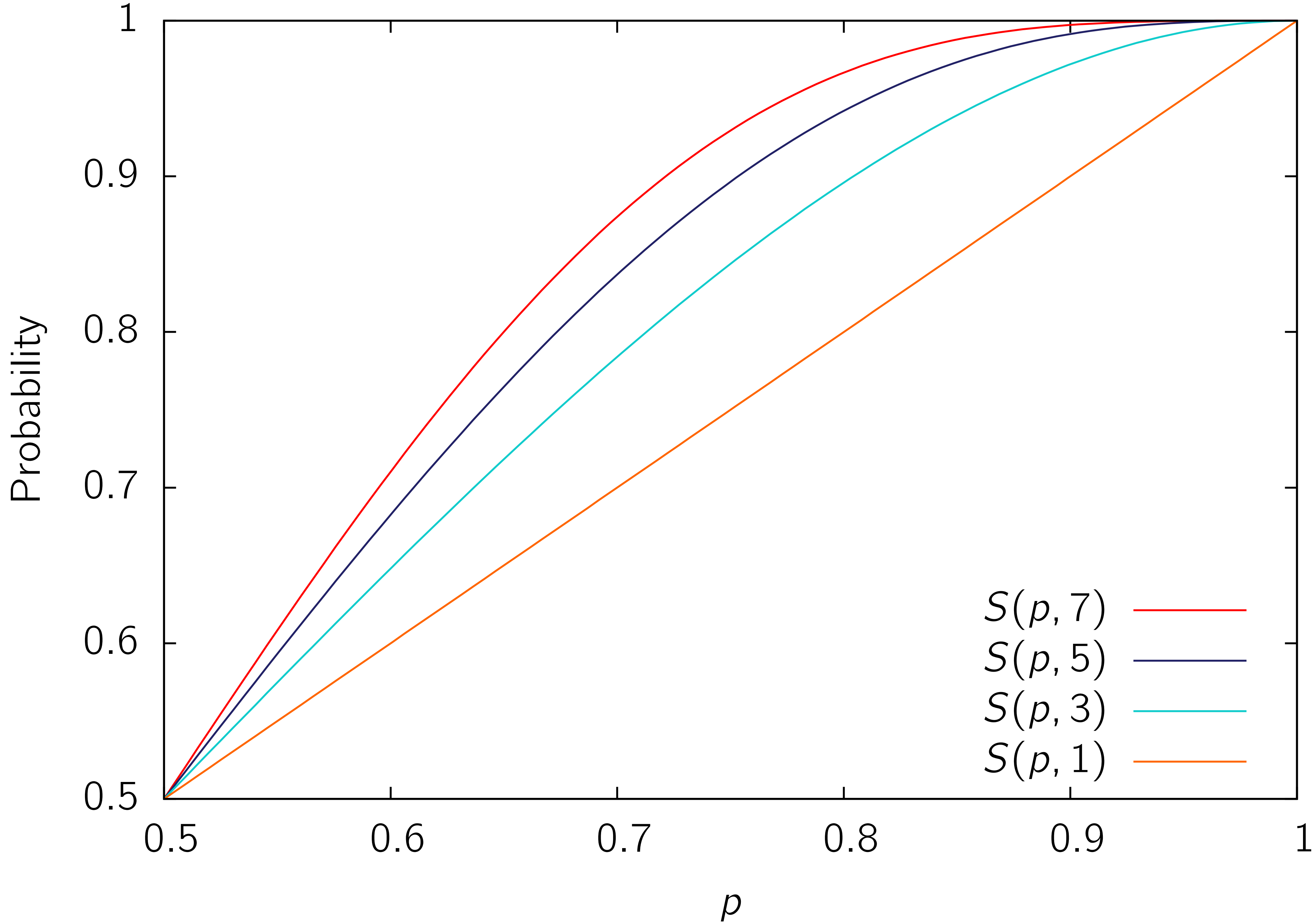
$$S(p,3) = p^3 + 3\,p^2(1-p) = p^2(3-2\,p)$$
$$= \left(\frac{1}{2} + \epsilon\right)^2 (2-2\,\epsilon)$$
$$= \frac{1}{2} + \frac{3}{2}\,\epsilon - 2\,\epsilon^2$$

- Increased chance better team winning:

$$S(p,3) - S(p,1) = \frac{\epsilon}{2} - 2\epsilon^3$$

- For p=0.51, so that ϵ = 0.01, this is 0.005

Amplification grows with series length



Exercise: calculate amplification factor for longer series lengths

What's an appropriate *p*?

- To answer the central question, Mosteller aims to estimate a value of *p* based on historical World Series data
- Examine period from 1905 to 1951. During this time, there were 44 seven-game series, plus 4 nine-game series
- Each World Series has one team from each league
  - National League (St. Louis Cardinals, S.F. Giants, *etc.*)
  - American League (Boston Red Sox, New York Yankees, *etc.*)

A.L. versus N.L. comparison

- First observation: the A.L. team wins more frequently, winning 31 out of 48 series
- Won 159 out of 275 games: 57.82%
- Has the A.L. team done significantly better than the N.L. team?
- Assume null hypothesis of $p = \frac{1}{2}$
- Probability of winning 31 or more = 0.0297

Mosteller's first estimation

- Assume that the better team has the same *p* each year
- Define $q = 1 - p$
- Analyze the 44 seven-game series:

TABLE 5

GAMES WON (SEVEN-GAME SERIES ONLY)

Winner	Loser	Frequency	Theoretical Proportion
4	0	9	p^4+q^4
4	1	13	$4p^4q + 4pq^4$
4	2	11	
4	3	11	
	Total	44	

Where does this come from?

Combinations of games won

- Four combinations for better team to win in a 4 – 1 series

WBBBB, BWBBB, BBWBB, BBBWB

- How many ways are there for a better team to win a 4 – 2 series?

3 B's and 2 W's in any order

*****B

$$\binom{5}{2} = \frac{5!}{2!\,3!} = 10$$

Mosteller's full table

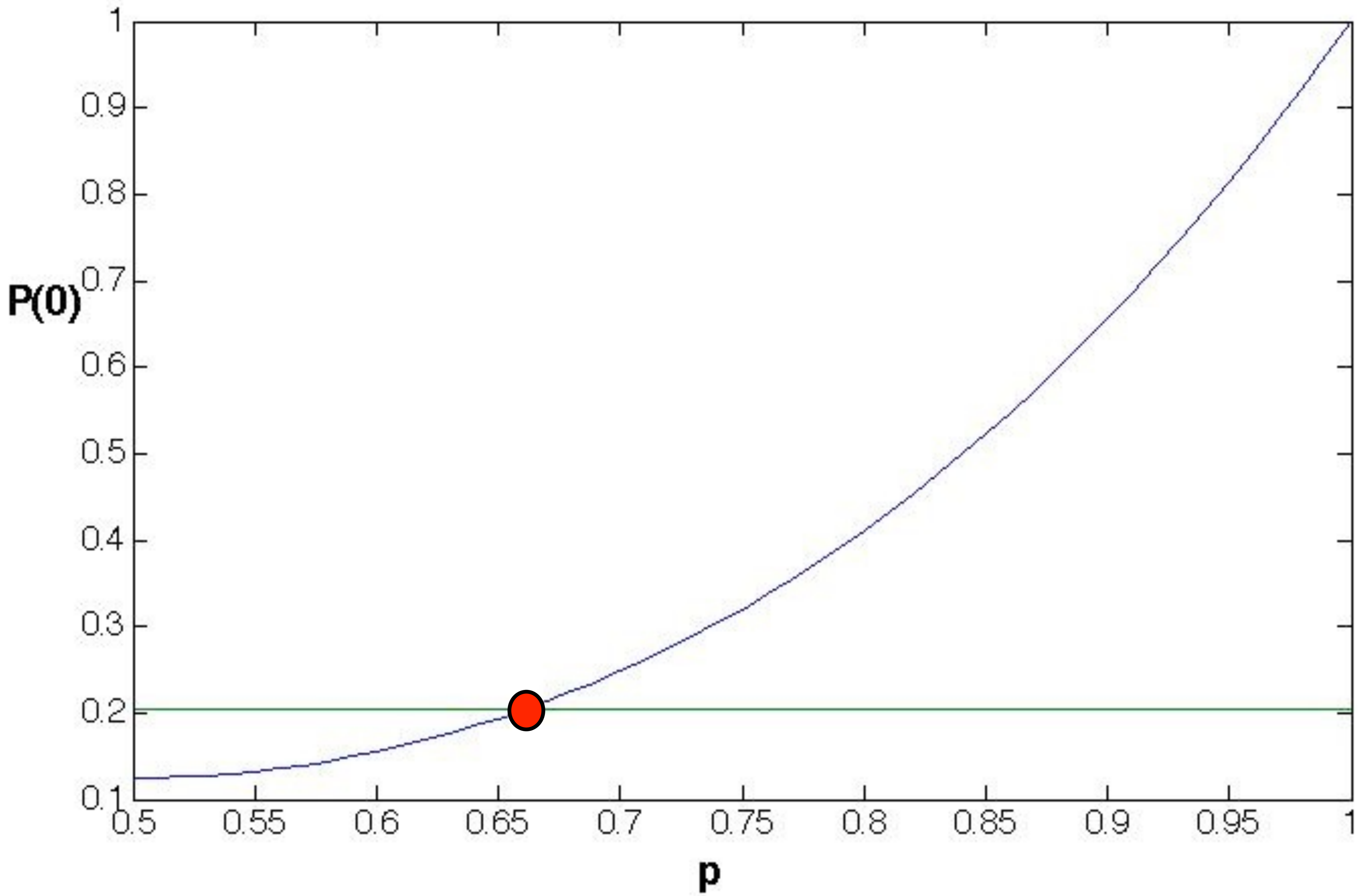
TABLE 5

GAMES WON (SEVEN-GAME SERIES ONLY)

Winner	Loser	Frequency	Theoretical Proportion
4	0	9	p^4+q^4
4	1	13	$4p^4q + 4pq^4$
4	2	11	$10p^4q^2+10p^2q^4$
4	3	11	$20p^4q^3+20p^3q^4$
	Total	44	1

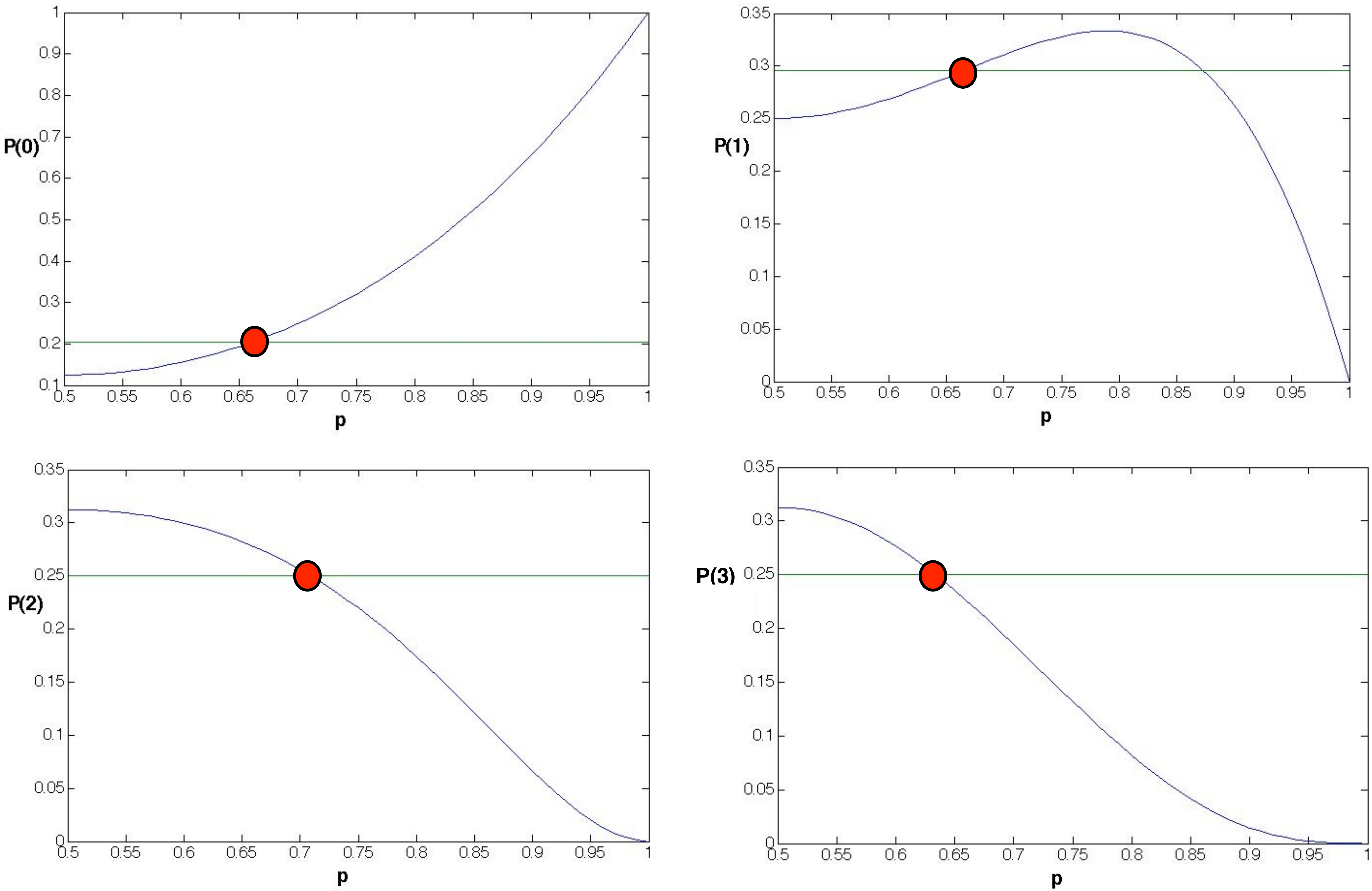
- Can try choosing *p* to fit the observed frequencies
- Let's just examine the 4–0 series, which occur 9/44 = 20.4% of the time

An estimate for *p*



Looks like $p = 0.65$ is pretty good

In fact $p = 0.65$ fits all the data



Mosteller's three estimation methods

- Mosteller considers three methods of systematically choosing *p*:
  1. He makes the average number of games lost by the winning team match
  2. He uses a maximum likelihood argument
  3. He uses a χ^2 statistic
- We'll examine the first two

Average number of losing games

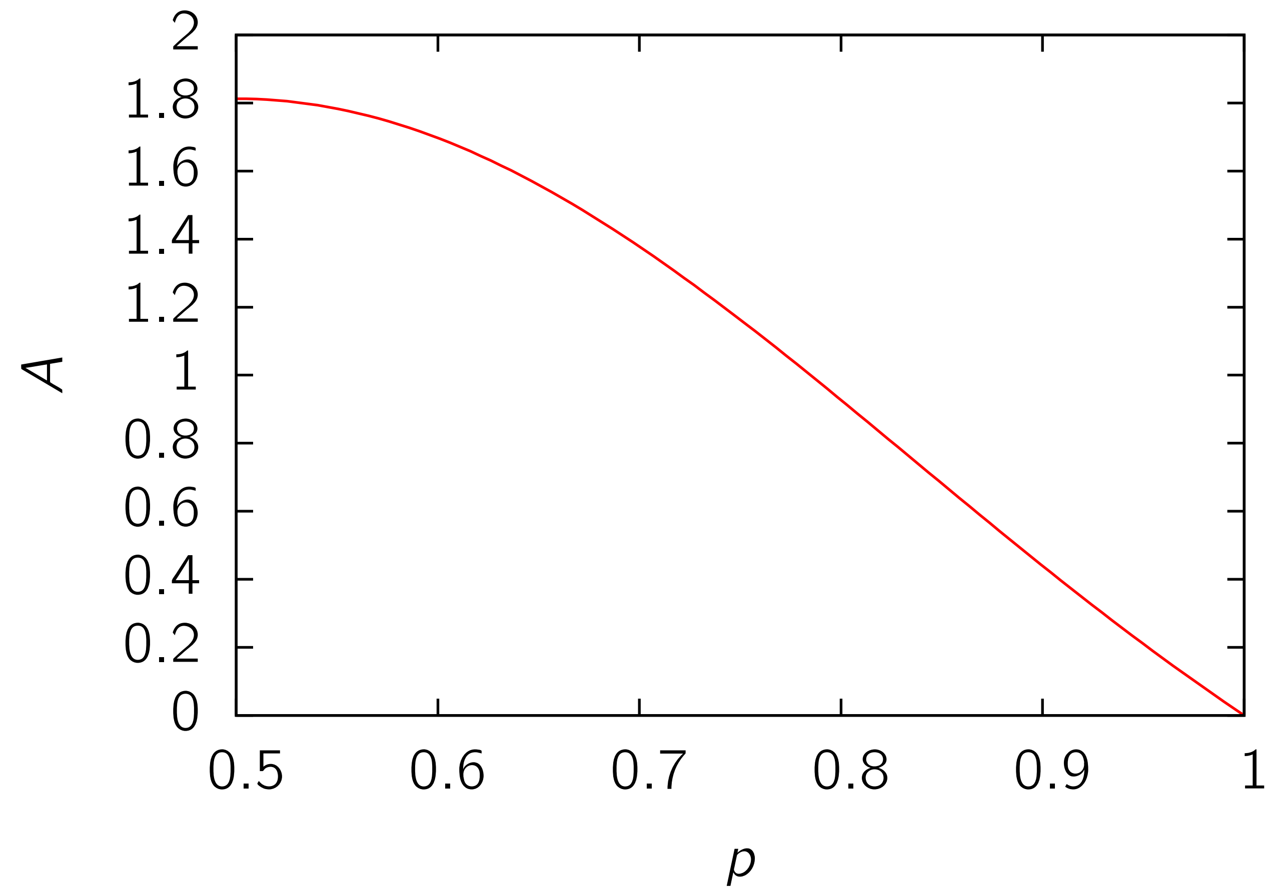
$$\begin{aligned} A &= \text{Average number of games won per Series by the Series-loser} \\ &= 1(4p^4q + 4pq^4) + 2(10p^4q^2 + 10p^2q^4) + 3(20p^4q^3 + 20p^3q^4) \\ &= 4pq[(p^3 + q^3) + 5pq(p^2 + q^2) + 15p^2q^2]. \end{aligned}$$

- **After some manipulations,**

$$A = 4pq[1 + 2pq + 5p^2q^2]$$

- **Problem: find value of *p* corresponding to the value in the data of *A* = 1.5455**

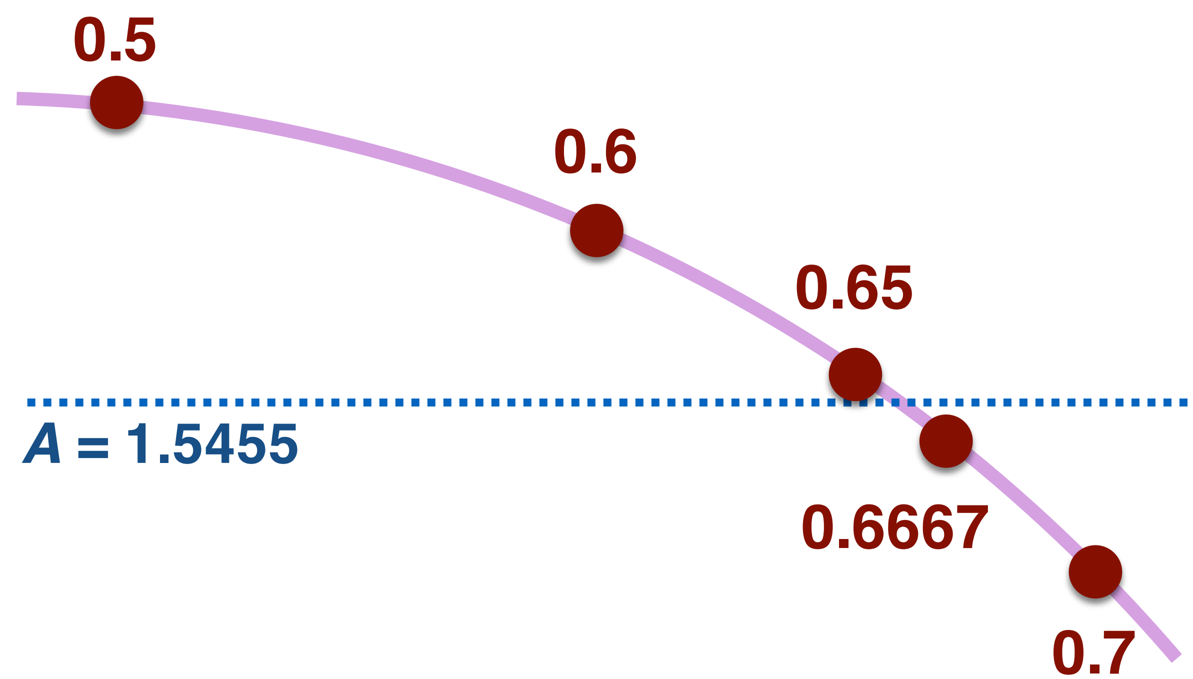
- **But the right hand side is a sixth order polynomial in *p***



Root finding the old school way

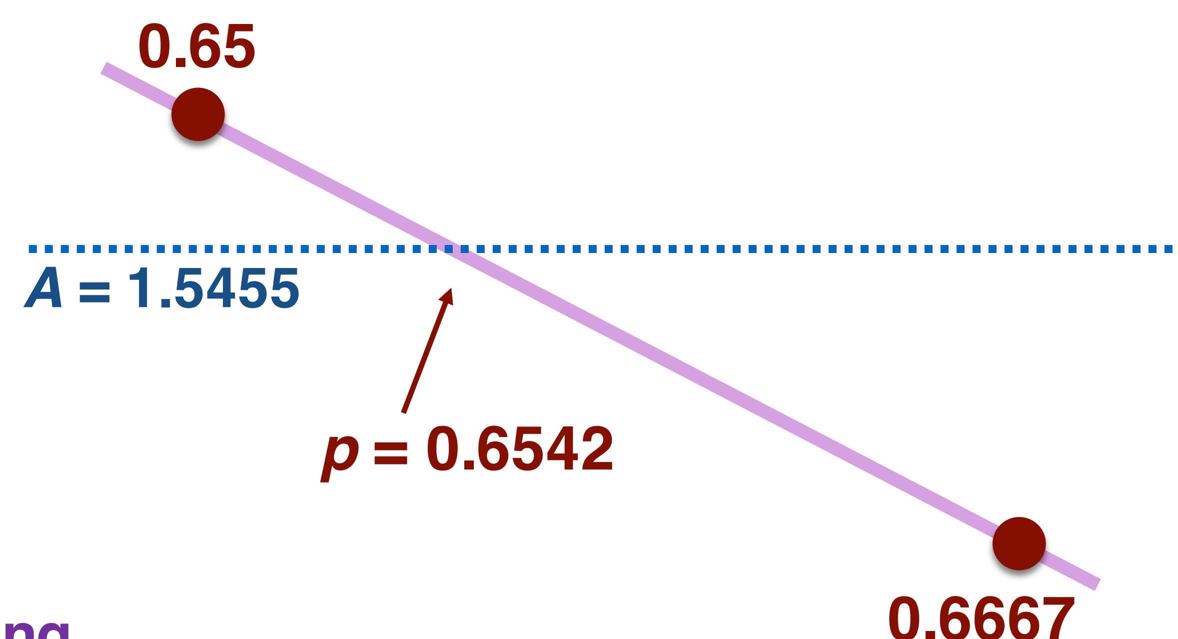
- **Mosteller calculates the value of *A* for a number of different values of *p*:**

p	A = Average Wins Expected by Loser
0.5	1.8125
0.6	1.6973
0.6500	1.5596
0.6667	1.5034
0.7	1.3780



- **He then linearly interpolates between the two**

- **Similar to modern computer root-finding strategies (see "bisection search")***



*See APMTH 111: Introduction to Scientific Computing

2. Maximum likelihood

Method 2. If $P(0)$, $P(1)$, $P(2)$, $P(3)$ are the probabilities that the Series-losing team wins 0, 1, 2, or 3 games respectively in a Series, then the maximum likelihood approach involves finding the value of p that maximizes

$$[P(0)]^9[P(1)]^{13}[P(2)]^{11}[P(3)]^{11}.$$

The numbers 9, 13, 11 and 11 are the frequencies tabulated in Table 5 and the $P(x)$ are given in algebraic form in the Theoretical Proportions column in Table 5. Although tedious, this maximization was done, and the estimate obtained was 0.6551, encouragingly close to that obtained from Method 1.

Summary of estimations of *p*

Method	*Estimate*
Average Wins by Series Loser	0.6542
Maximum Likelihood	0.6551
Minimum Chi-square	0.6551

- **All three approaches give almost the same answer, which is very close to our initial guess of $p = 0.65$**
- **Can now answer the original question of how often the better team wins:**

$$S(0.65, 7) = 0.80$$

- **This means that 20% of the time, the better team doesn't win**

How often does the A.L. have the better team?

- Recall that the A.L. team won 31 out of 48 times
- Let x be the proportion of series where the A.L. has the better team. Then

$$0.80x + 0.20(1 - x) = \frac{31}{48} = 0.646$$

- After some algebra,

$$\begin{aligned} 0.60x &= 0.446 \\ x &= 0.743. \end{aligned}$$

A key assumption in the analysis

- Same p for all world series
- More realistic to say that the better team will have a varying advantage. Sometimes one team may be much better, sometimes both teams may be almost equally matched

Scenario 1
p=0.7 for all games

$S(0.7,7)=0.87$

Scenario 2
p=0.5 for half, p=0.9 for half

$S(0.5,7)=0.5$, $S(0.9,7)=1.00$

Average $S = 0.75$

- See Mosteller's paper for a more advanced approach with varying p

Tests of binomial assumptions

- **Mosteller acknowledges the limitations of assuming binomial trials with a fixed *p* throughout a series**
- **Addresses issues mentioned in class (home versus away games, “momentum” gained after winning)**
- **In addition, more recent data shows a large increase in 4–3 games!**
- **Why is this? Examine in the first computer lab**

Different ways to model the world series

1. Make a physical simulation of the entire ballpark and every single play, modeling the way the ball moves through the air, and the expected movement of players.
2. Compile detailed stats on each player and make a probabilistic simulation of every single ball played.
3. Treat each game as a flip of an unequal coin with varying *p*, taking into account factors like home/away game, team strength, *etc.*
4. Treat each game as a flip of an unequal coin with constant *p.* (Mosteller)
5. A world series competition is just a single coin toss.

A variety of modeling choices

Accuracy

Complexity

Too many unknown parameters

Too difficult to solve

Simplicity

Easier to calculate

Not as realistic

Provides more insight

1 2 3 4 5

Mosteller's choice

Choosing the right level of detail in a mathematical model is a crucial skill, and involves many trade-offs. It depends on what the purpose of the model is. Mosteller's choice of model complexity is guided by his original question: "how often does the better team win?" The level of detail in his model is just enough to answer this question.

Break

Getting started with Matlab

- Tips on learning for beginners:
  - There’s no substitute for trying it yourself
  - There’s extensive online documentation,e.g, https://www.mathworks.com/help/matlab/
  - Come and see us
  - If stuck on a specific issue, try Googling it

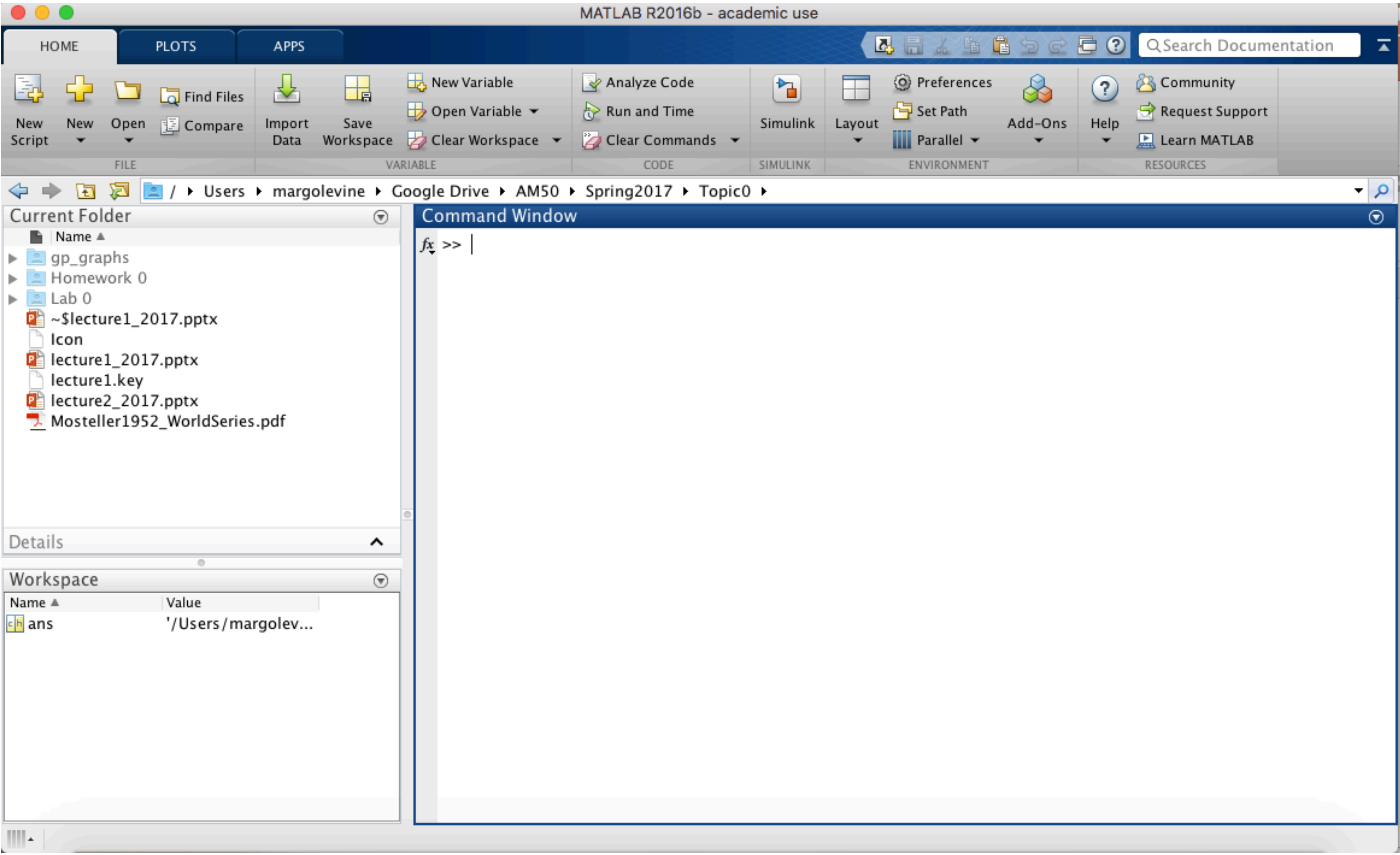
Getting started with Matlab

- Download and install: http://downloads.fas.harvard.edu/download
- Ziwei's section on Monday (time TBD)
- Matlab Bootcamp, starting Monday evening (https://wiki.harvard.edu/confluence/display/fasmatlab/MATLAB+Boot+Camp+Home)
- Matlab on-ramp (http://www.mathworks.com/academia/student_center/tutorials/mltutorial_launchpad.html)

Computer lab 0

- On your own or in Ziwei's section on Monday
- Set up Matlab and learn basic commands
- Run a small program to simulate Mosteller's model. We'll give you the program but you need to explain what it does
- Look at the output
- Try making some small modifications (*e.g.* varying *p* for each game)

Workspace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Mosteller1952_WorldSeries.pdf
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fx >>

Hello World!

Input program

```
disp('Hello World!')
```

Output

Hello World!

Example: hello_world.m

http://en.wikipedia.org/wiki/Hello_world

Variables and basic math operations

Input program

```
%Create two variables
% and do some arithmetic
a = 1;
b = 2*a;
a = a+1;

% Print the contents of the
% variables
a
b
```

Output

```
a =
        2
b =
        2
```

Notes:

- **Comments in Matlab start with a "%" symbol**

Example: variables.m

Arrays

Input program

```
%Create an array, append an item,
% and demonstrate various outputs
a = [6,3,1,4];
b = 10;
a = [a, b];

a(1)
a(1:3)

%Create an matrix, append an array,
% and demonstrate various outputs
c = [1 2; 3 4];
d = [0 0];

c_vert = cat(1, c, d)
c_horz = cat(2, c, d')
```

Output

```
>> arrays

ans =

     6


ans =

     6     3     1


c_vert =

     1     2
     3     4
     0     0


c_horz =

     1     2     0
     3     4     0
```

Example: arrays.m

Conditions and “if” statements

Input program

```
% clears all the previously defined
% variables
% closes all previous figures
clear all;
close all;

year = 2;
name = 'Margo';

% Do different things depending on
% the condition
if strcmp(name, 'Eleanor') == 1
    year = 2014
elseif strcmp(name, 'Sasha') == 1
    year = 2012
else
    year = 0
end
```

Output

```
>> conditions

year =

     0

>>
```

Example: conditions.m

Program flow and loops

Input program

```
%create an array 1, 2, ..., 10
N = 10;
for i = 1:N
    a(i) = i;
end

%do it another way
i = 1;
while i <= 10
    b(i) = i;
    i = i + 1;
end

%do it without a loop
c = 1:N;

a
b
c
```

Output

```
>> loops

a =

     1     2     3     4     5     6     7     8     9    10


b =

     1     2     3     4     5     6     7     8     9    10


c =

     1     2     3     4     5     6     7     8     9    10
```

Example: loops.m

Next week: epidemiology

- Using search engine queries to detect flu epidemics
- Paper will be posted on Canvas site, and will be discussed on Tuesday